CollabCRM People Training Module - Flow & Field Specification

Document Overview

This document specifies the complete user navigation flows, screen transitions, form fields, allowed values, validation rules, and behavioral rules for the **CollabCRM People & Training Module**. Every screen, field, allowed value, and interaction described below matches the live system.

SECTION 1: SYSTEM NAVIGATION & SCREEN FLOWS

Overall Navigation Flow

1. **Application Root (CollabCRM Enterprise Suite)** opens to the **Top Global Navigation Bar** and the **Left Collapsible Navigation Drawer**.
2. **Role Switching Flow (Top Navigation)**:
   - Users can switch between the **Admin / HR Operations View** and the **Employee Learning Portal**.
   - Selecting Employee View navigates directly to **My Training**, where learners open the **3-Tab Course Learning Portal** (`1. Overview`, `2. Learning Resources`, `3. Assessment & Questions`).
3. **Left Navigation Drawer Flow**:
   - **Employees**: Opens the Employee Directory, Employee Profile (with `Skills` section showing Training sources & `Skill Award History` modal, `Training History` tab, and `My Certificates` tab), and the Add Employee modal.
   - **Training Hub**: Provides access to Dashboard, Training Management (Training Catalogue), Assignments, Training Progress, and Certificates.
   - From **Training Management**: Users can open **Create / Edit Training** (including **3. Training Recognition** and **4. Assessment & Questions**) or click View to open **Training Detail** (General Info with `Training Recognition` summary card, Curriculum & Modules, Enrolled Employees, Status History).
   - From **Assignments**: Users can open the **Assign Training Slideover Modal** or click any record to open **Assignment Details** (with **View Assessment** review support).
   - From **Certificates**: Users can expand employee rows and open the **Verified Certificate Modal** (signed by Khushi Bhatt, Head of HR).

- **Config Hub**: Provides quick navigation to **Training Categories** (master category table & Add Category modal), **Skills & Skill Levels** (`Config > Skills` master skills table, `+ Add Skill` modal, active/inactive toggle, and configured skill level hierarchy), and **Roles & Permissions** (Roles Table, Add Role modal, and Role Permission Matrix).

Flow 1: Top Navigation & Role Switching

- **Global Header**: Contains company branding (CollabCRM People), global search (Search employees, programmes), notifications bell, and the Active User Profile indicator (`Tanya Kapoor - Business Analyst • Employee` or `Ishdeep Anand - Admin • Management`).
- **Role Switching**: Clicking the user badge switches between **Admin/HR View** and **Employee User Role View**. Switching to an Employee role redirects directly to the Employee Dashboard and personal learning workflows.

Flow 2: Navigation Drawer Structure

The left collapsible sidebar provides access to:

1. **Employees**: Complete employee directory with profile cards and the Add Employee modal.
2. **Training Hub**:
   - **Dashboard**: Real-time metrics (Active Learners, Total Assignments, Completion Rate, Average Score), urgent review queue, and performance analytics.
   - **Training Management (Training Programmes)**: Full course catalogue, publication status filters (`All` , `Published` , `Draft`), search, `Status ↕` column (`Published` / `Draft` badges), View & Edit actions, and `+ Add Training` action. (`Archived` status has been removed; once published, only `Active` / `Inactive` toggling can be performed inside Programme Detail.)
   - **Assignments**: Central assignment register with the **ASSIGNED SCOPE** column (Employee, Department, Designation), **41 Total** assignment count, and 5-page pagination (10 items per page). Clicking a record opens **Assignment Details**.
   - **Progress**: Dual-mode tracking (1. Employee-wise Training Completion vs 2. Departmental Compliance Audit), 4 summary cards (Active Learners, Total Assignments, Completed, Avg Score), and CSV export.
   - **Certificates**: Expandable master-detail table defaulting to **collapsed** employee rows with expand/collapse chevrons, plain-text earned programmes, clickable certificate preview, and a right-aligned **Actions** column directly under **All Records**.
3. **Config Hub**:
   - **Config Flyout**: Compact left drawer panel offering quick access to **Training Categories**, **Skills**, and **Roles**.

- **Training Categories**: Master category table and **Add Category** action button placed directly in the top **Training Config** banner header (without extra sub-tab or duplicate table card header rows).
- **Skills & Skill Levels (`Config > Skills`)**: Master Skills table (`Skill Name`, `Description`, `Status`, `Created By`, `Actions`), `+ Add Skill` drawer/modal (with duplicate-name validation), active/inactive status toggling, and configured ordered **Skill Levels** progression strip (`Beginner (Level 1)`, `Intermediate (Level 2)`, `Advanced (Level 3)`).
- **Roles & Permissions**: 11-role access control table (Sr. Business Analyst, Senior Recruiter, CEO, Project Manager, BDE, Admin/HR, Employee, Team leader, Management, Recruiter, BDM), `+ Add Role` slideover modal, and interactive Role Permission Matrix.

(Note: The legacy Reports drawer item has been removed.)

Flow 3: Training Management (Create / Edit Programme Flow)

- Accessed via **Training Management > + Add Training** or **Edit Programme**.
- Features an **Auto-Fill Demo** button to populate sample data.
- Structured across four core sections:
  1. **Core Training Details**: Training Name, Training Category, Difficulty Level, Duration, Learning Objectives, and **Completion Requirements** (multi-select checkbox dropdown with options `Watch all required videos`, `Pass the assessment`, `Submit any required documents`, and `Get approval from the manager`; checkboxes toggle directly without `Clear` / `Done` buttons and the menu closes on outside click).
  2. **Learning Resources Builder**: Dynamic resource list (Video/YouTube, PDF Document, External Link, Article).
  3. **Training Recognition**: Independent toggles for **Certificate Generation** (default OFF) and **Add Skill to Employee Profile** (default OFF), plus **Require Reviewer / Manager Approval Before Completion** (default OFF). When **Add Skill to Employee Profile** is ON, displays multi-select **Select Skills** (populated live from active `Config > Skills`, displaying the selected skill names directly inside the dropdown box and as removable chips, with an inline **+ Create New Skill in Config** action) and **Skill Level** (populated from configured Skill Levels). Positioned above Section 4 so chosen skills drive the Assessment Question List.
  4. **Assessment & Questions Builder**: Enable Assessment toggle, Assessment Title, Passing Score (`Passing Score (%) *`), Max Assessment Attempts, Assessment Description, **Total Marks** numeric text field (`Total Marks:`, auto-synced when questions are generated from selected skills), **↻ Sync Skill Questions** button (when skills are selected), and dynamic multi-format question builder that automatically populates & maintains skill-aligned questions (`Skill: <Name> · <Level>`) from selected Skills & Skill Level while preserving manually added questions and supporting manual marks, live mismatch validation against `Total Marks`, and bordered SVG-icon **Duplicate** / **Delete** buttons.

- Header Actions: **Auto-Fill Demo**, **Cancel**, **Publish** (or **Update & Publish** / **Save Changes**), and text-only **Save as Draft** dropdown option (no floppy disk icon).

Flow 4: Assign Training Slideover Modal Flow

- Accessed via **+ Assign Training** button in Header, Training Programmes, or Assignments table.
- Configurable targeting parameters:
  - Bound **Training Programme** display field (shows selected or first programme).
  - Multi-select **Department** filter with **Select All / Deselect All** and dynamic count.
  - Multi-select **Designation** filter with **Select All / Deselect All**.
  - Multi-select **Employee (s)** list with **Select All / Deselect All** shortcut and reactive count.
  - Mandatory **Start Date** and optional **End Date**.
- Action Buttons: **Auto-Fill**, **Cancel**, and **Assign Training**.

Flow 5: Certificates Verification Flow

- **Master-Detail Expandable Table**:
  - Defaults to **collapsed** employee rows; expanding a row reveals child sub-rows for additional certificates earned by that employee.
  - Master row displays Employee Name, Department/Role, Primary Earned Programme, and an expand/collapse toggle chevron.
  - Plain-text format for programme names (no dropdowns or passed badges inside the cell). Clicking a certificate card or the view icon opens the verified certificate modal.
  - **Right-Aligned Actions**: The **Actions** column header and eye icon buttons are aligned to the far right edge, positioned directly below the **All Records** category filter.
- **Certificate Modal**:
  - Displays the official certificate of appreciation with centered **CollabCRM** brand header, gold seal icon, and dark-navy corner accent.
  - Dynamically bound to the exact completed programme and learner name.
  - **Authorized Signatory**: Configured strictly as **Khushi Bhatt, Head of HR**.

Flow 6: Employee User Role Learning Portal Flow

- Accessed by switching user to **Tanya Kapoor** or navigating to **Training > My Training**.

- **Dynamic State Integration**: Newly created and published programmes from Admin/HR view automatically populate in the employee's training list. **Draft programmes are excluded from all employee views** (Employee Dashboard, My Training, Employee Course Detail, and Employee Profile Training History).
- **Inactive Programme Read-Only Handling**:
  - Programmes marked **Inactive** (including programmes **Completed** first and then set to **Inactive**) remain visible in **My Training** with an **Inactive** status pill.
  - Opening an **Inactive** programme shows an **Inactive warning banner** and disables all actions inside (Start Learning, Mark Complete, external resource links, assessment inputs/selects/checkboxes/radios, Submit Assessment, Retake Assessment, and certificate download).
- **Training History (Employee Profile > Training History)**: Displays enrollment metrics (Total Enrolled, Completed & Passed, In Progress, Certificates Earned), a compact **+ Assign New Training** button, and assigned training records.
- **3-Tab Course Learning Layout**:
  - **Tab 1 — 1. Overview**: Progress bar, completed modules counter, course summary, learning objectives, configured **Completion Requirements** checklist (e.g., Watch all required videos, Pass the assessment, Submit any required documents, Get approval from the manager), and the **Training Recognition** summary card showing configured recognition rewards (Certificate / Awarded Skills + configured Skill Level) and status (`Awarded Upon Completion` , `Pending Reviewer Approval` , or `Recognition Awarded ✓`). Once all required steps and optional reviewer approval are satisfied, the training is marked **Completed**.
  - **Tab 2 — 2. Learning Resources**: Resource checklist with external link openers and completion toggles (disabled if programme is Inactive).
  - **Tab 3 — 3. Assessment & Questions**:
  - **Header**: Clean title **3. Assessment & Questions** (without any Unlocked tag).
  - **When Assessment Toggle is OFF**:
  - Inside Tab 3: Renders a clean empty container ("No Assessment Configured — Assessment is disabled for this training programme. Learners achieve completion upon completing all configured requirements"). No questions, no scoring inputs, and no submit button are shown.
  - Left Sidebar: Passing Assessment Score displays **N/A (Assessment Disabled)**. Action button displays **Start Learning Modules →** or **Training Completed ✓** (never prompts to take an assessment).
  - Tab 2 Navigation: Bottom button displays **All Modules Completed ✓**.
  - **When Assessment Toggle is ON**:
  - Inside Tab 3: Renders the exact question types, marks, and options configured by Admin/HR (without per-question Mandatory checkboxes or labels).

- Left Sidebar: Passing Assessment Score displays **80%** (clean percentage without extra strictly-greater-than hint). Action button displays **Take Assessment →** (or **Download Certificate** when Certificate Generation is enabled and all required steps are complete; hidden when Certificate Generation is OFF).
- Result & Recognition Cards: Displays unified **Completed** badge when passed, plus **Skills Added to Your Profile** purple summary card when skills are awarded.
- **View Assessment (HR / Manager Assignment Detail)**: Even when an employee completes or fails an assessment (after attempts are submitted or exhausted), HR and Reporting Managers can open **View Assignment > View Assessment** to inspect submitted answers and evaluation status.
- **Employee Profile > Skills Section**: Reuses the existing **Skills** block inside the Employee Profile left sidebar card (no separate Training Skills tab). Displays skill name, current Skill Level badge, Training source tag (when acquired via Training), clickable **Skill Award History** modal showing progression & Training sources, and authorized-only `+ Manage` skill modal (Admin/HR/Reporting Manager only; Employees cannot self-award or modify skills).

SECTION 2: FORM FIELD VALUES & VALIDATION SPECIFICATIONS

1. Create Training Programme Form (Training Management > Create Programme)

Section 1: Core Programme Details

- **Training Name:**
  - Mandatory: Yes (`*`)
  - Type: Alphanumeric Text Input
  - Max. Length: 120
  - Min. Length: 3
  - Default value: Empty
  - Validation: Cannot be blank or whitespace-only. Must be unique across active programmes.
- **Training Category:**
  - Mandatory: Yes (`*`)
  - Type: Dropdown (Single Select)
  - Default value: Recruitment & HR Operations
  - Allowed Values:
  - Recruitment & HR Operations
  - Engineering & Tech Architecture

    - Sales & Business Development
    - Governance & Compliance
    - Product & Design
- **Difficulty Level:**
    - Mandatory: Yes (`*`)
    - Type: Dropdown (Single Select)
    - Default value: Intermediate
    - Allowed Values:
    - Beginner
    - Intermediate
    - Expert
- **Duration *(`Expected Duration (Hours)`):***
    - Mandatory: Yes (`*`)
    - Type: Numeric (Integer)
    - Max. Value: 200
    - Min. Value: 1
    - Default value: Empty (Placeholder: `e.g. 4`)
    - Validation: Positive integer; decimals or negative values not permitted.
- **Completion Requirements:**
    - Mandatory: No (Optional)
    - Type: Multi-select Checkbox Dropdown (closes on outside click or header re-click; no `Clear` or `Done` footer buttons)
    - Default value: Empty (Placeholder: `Select completion requirements...`)
    - Allowed Values:
        1. `Watch all required videos`
        2. `Pass the assessment`
        3. `Submit any required documents`
        4. `Get approval from the manager`
    - Behavior: Controls what the employee must complete to finish the training. Once all selected required steps are completed, the training status transitions to **Completed**.

Section 2: Learning Resources (YouTube, External Courses, Websites, Videos, PDFs, Docs) *

- **Resource Type:**

- Mandatory: Yes
- Type: Dropdown (Single Select)
- Default value: Video / YouTube
- Allowed Values:
- Video / YouTube
- PDF Document
- External Link
- Article

- **Resource Title:**
  - Mandatory: Yes
  - Type: Alphanumeric Text Input
  - Max. Length: 150
  - Min. Length: 2
  - Default value: Empty in newly added rows (Pre-populated starter rows: "Module 1: Architecture Video Lecture", "Module 2: Implementation Playbook & Guidelines")
  - Validation: Clear module identifier required.
- **Resource URL:**
  - Mandatory: Yes
  - Type: URL Input
  - Max. Length: 500
  - Min. Length: 8
  - Default value: Empty in newly added rows (Placeholder: `https://...`)
  - Validation: Must be a valid web link starting with http:// or https://.

Section 3: Training Recognition (Create / Edit Training)

- **Certificate Generation:**
  - Mandatory: No
  - Type: Toggle Switch (`ON` / `OFF`)
  - Default value: `OFF` (`false`)
  - Behavior: Independent toggle allowing Admin/HR/Reporting Managers to enable or disable automatic certificate generation per Training. When `OFF`, no certificate is issued and certificate download prompts are hidden upon completion. When `ON`, uses the verified certificate issuance workflow (`TRN-YYYY-Serial`) upon verified completion.

- **Add Skill to Employee Profile:**
  - Mandatory: No
  - Type: Toggle Switch (`ON` / `OFF`)
  - Default value: `OFF` (`false`)
  - Behavior: Independent toggle. When `OFF`, skill configuration fields remain hidden and no skills are awarded on completion. When `ON`, renders **Select Skills ** *and* ***Skill Level*** **|**. Both Certificate Generation and Add Skill to Employee Profile work independently (Certificate Only, Skill Only, Both, or Neither). Positioned directly above **Section 4: Assessment & Questions** so selected skills drive automatic question generation.
- **Select Skills (Displayed when `Add Skill to Employee Profile` is ON):**
  - Mandatory: Yes (`*` , when `Add Skill to Employee Profile` is `ON`)
  - Type: Searchable Multi-select Checkbox Dropdown (with removable skill chips & `Done` close button)
  - Display Format: Trigger input directly shows the comma-separated selected skill name(s) (e.g., `Account Management, React`) with overflow ellipsis + full tooltip instead of a generic count string.
  - Data Source: Loaded dynamically from **active** skills in `Config > Skills` (`GET /api/config/skills`); never hardcoded.
  - Inline Creation: Authorised users can click **+ Create New Skill in Config** directly inside Section 3 to add a new skill via `Config > Skills` ; saving automatically selects and makes the skill available in the Training skill dropdown without leaving the form.
  - Validation: At least 1 active skill must be selected; duplicate selections within the same Training are prevented; inactive skills from `Config > Skills` are rejected on new configurations.
- **Skill Level (Displayed when `Add Skill to Employee Profile` is ON):**
  - Mandatory: Yes (`*` , when `Add Skill to Employee Profile` is `ON`)
  - Type: Dropdown (Single Select)
  - Allowed Values: Loaded dynamically from configured Skill Levels in `Config > Skills` (`Beginner` , `Intermediate` , `Advanced` with ordering `1 < 2 < 3`). No separate skill-level master is created. Changing the selected `Skill Level` dynamically updates auto-generated skill questions in Section 4 to match the chosen proficiency level.
  - Validation: Must select a valid configured skill level.
- **Require Reviewer / Manager Approval Before Completion:**
  - Mandatory: No
  - Type: Toggle Switch (`ON` / `OFF`)
  - Default value: `OFF` (`false`)

- Behavior: When `ON` (or when `Get approval from the manager` is selected in Completion Requirements), verified completion enters `Under Review` (`managerApprovalStatus: Pending Approval`) and holds Certificate Generation & Skill Assignment until an authorised Admin, HR, or Reporting Manager approves via `Approve Completion & Issue Recognition` (`POST /api/training/approve-completion`). Employees cannot approve their own completion (`HTTP 403`).

Section 4: Assessment & Questions Builder

- **Enable Assessment Toggle:**
  - Mandatory: Yes
  - Type: Toggle Switch (Active / Inactive)
  - Default value: Active
- **Skill-Based Automatic Question Generation & Sync (`buildSkillBasedQuestions`):**
  - When **Add Skill to Employee Profile** is `ON` and skills are selected in **Section 3: Training Recognition**, **Questions List** automatically populates skill-aligned graded questions tailored to each selected skill (`Skill: <Name> · <Level>`) and the chosen **Skill Level** (`Beginner` , `Intermediate` , `Advanced`), setting clean assessment defaults (`Final Training Assessment` , Passing Score `80` , Max Attempts `3`) when blank.
  - Supports curated question banks for all configured domain skills as well as dynamic contextual question generation for any newly created custom skill in `Config > Skills` .
  - Removing or adding skills updates only auto-generated skill questions while preserving manually added or duplicated custom questions (`+ Add Question` / **Duplicate**). Users can also click **↻ Sync Skill Questions** anytime to refresh skill-aligned questions.
- **Assessment Title:**
  - Mandatory: Yes (`*` , when Enable Assessment is Active)
  - Type: Alphanumeric Text Input
  - Max. Length: 120
  - Min. Length: 3
  - Default value: Final Training Assessment (auto-populated if blank when skill questions sync)
- **Passing Score (%):**
  - Mandatory: Yes (`*` , when Enable Assessment is Active)
  - Type: Numeric (Integer)
  - Max. Value: 100
  - Min. Value: 1

  - Default value: Empty on create (`Auto-Fill Demo` or skill-based sync populates `80` when blank)
  - Validation: Benchmark passing percentage (clean label `Passing Score (%) *` without extra `(Strictly > 80% to pass)` helper text).
- **Total Marks (`Questions List` Header):**
  - Mandatory: Yes (when Enable Assessment is Active)
  - Type: Numeric Text Input (`inputMode: numeric`)
  - Default value: Empty (`""` , placeholder: `e.g. 100` ; automatically synchronized to the sum of question marks whenever skill-based questions are auto-generated/synced, displaying `Matched ✓` , while preserving manual overrides)
  - Validation: Must be greater than `0` ; if all added questions have marks `> 0` and `questionsMarksSum !== Number(Total Marks)` , an inline error (`Total question marks (X) do not match Total Marks (Y).`) is displayed and save/publish is blocked. When equal, a `Matched ✓` pill is displayed.
- **Max Assessment Attempts:**
  - Mandatory: Yes (`*` , when Enable Assessment is Active)
  - Type: Numeric (Integer)
  - Max. Value: 10
  - Min. Value: 1
  - Default value: 3
- **Assessment Description:**
  - Mandatory: No (Optional)
  - Type: Multi-line Textarea
  - Max. Length: 500
  - Default value: Complete the following questions to evaluate your understanding of this training.
- **Question Statement:**
  - Mandatory: Yes (`*`)
  - Type: Alphanumeric / Multi-line Textarea
  - Max. Length: 1000
  - Min. Length: 5
  - Default value: Empty for manual questions; populated contextually for skill-based questions
- **Question Type:**
  - Mandatory: Yes (`*`)
  - Type: Dropdown (Single Select)

    - Default value: Single Selection (Radio) for newly added questions
    - Allowed Values:
    1. Short Text Answer
    2. Long Text Answer
    3. Single Selection (Radio)
    4. Multiple Selection (Checkbox)
    5. Dropdown Selection
    6. True / False
- **Marks (Per Question):**
    - Mandatory: Yes
    - Type: Numeric Text Input (`inputMode: numeric`)
    - Min. Value: 1
    - Default value: `5` per auto-generated skill question; empty (`""` , placeholder: `0`) for manual `+ Add Question` entries
    - Validation: Must be greater than 0; sum of all question marks must equal configured `Total Marks` .
- **Mandatory Question:**
    - Mandatory: Yes
    - Type: Checkbox / Toggle
    - Default value: Checked / Active
- **Question Options & Correct Answers:**
    - **Single Selection (Radio), Multiple Selection (Checkbox), and Dropdown Selection**: Minimum 2 options required, each with Option Text. Single selection and dropdown require one selected Correct Answer; multiple selection requires at least one selected Correct Answer.
    - **Short Text Answer (`Configured Accepted Answers (Evaluated case-insensitively with trimmed whitespace):`)**: Minimum 1 accepted keyword/phrase required (`Enter accepted correct answer...` , without extra example hints).
    - **True / False**: Clean `Answer Option:` / Correct Answer selection (`True` or `False`).
    - **Long Text Answer**: Expected Answer / Evaluation Reference summary for manual reviewer grading.
    - **Question Actions**: Clean bordered **Duplicate** and **Delete** pill buttons with SVG icons (`8px` gap). Duplicating a skill-generated question creates a custom copy preserved across skill updates. (Question reorder buttons are removed.)
- **Persistence:**

- Persists `certificateGeneration`, `addSkillToProfile`, `selectedSkillIds`, `skillLevel`, and `reviewerApprovalRequired` across Create, Edit, Save as Draft, Publish, and full page reload (`POST /api/training/programmes` & `GET /api/training/programmes`).

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2. Assign Training Form (Assignments > + Assign Training)

- **Training Programme:**
  - Mandatory: Yes (`*`)
  - Type: Read-only Programme Display Box (bound to selected or first programme)
  - Default value: First programme (e.g. CollabCRM Recruitment & Client Portal Workflows)
- **Department:**
  - Mandatory: No (Optional targeting filter)
  - Type: Multi-select Dropdown with Select All / Deselect All
  - Default value: All Departments
  - Allowed Values: Business Analysis, Game, Collab, Web, Data analytics (plus dynamic departments from employee records)
- **Designation:**
  - Mandatory: No (Optional targeting filter)
  - Type: Multi-select Dropdown with Select All / Deselect All
  - Default value: All Designations
  - Allowed Values: Executive, Sr. Executive, Business analyst, Team leader, Manager, AVP (plus dynamic designations from employee records)
- **Employee (s):**
  - Mandatory: Yes (`*`)
  - Type: Multi-select Dropdown with Select All / Deselect All
  - Default value: Primary selected employee (first employee in directory)
  - Validation: Must have at least 1 employee selected before assignment submission. Displays reactive selected count.
- **Start Date:**
  - Mandatory: Yes (`*`)
  - Type: Date Picker
  - Default value: 2026-11-04
  - Validation: Required.
- **End Date:**
  - Mandatory: No (Optional completion deadline)

    - Type: Date Picker
    - Default value: 2026-12-01
    - Validation: Must be on or after Start Date.
- **Form Actions:** Auto-Fill, Cancel, Assign Training.

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3. Add Employee Form (Employees > + Add New Employee)

- **First Name:**
    - Mandatory: Yes (`*`)
    - Type: Text Input (Alphabetic)
    - Max. Length: 50
    - Min. Length: 2
    - Default value: Empty
- **Last Name:**
    - Mandatory: Yes (`*`)
    - Type: Text Input (Alphabetic)
    - Max. Length: 50
    - Min. Length: 2
    - Default value: Empty
- **Company Email:**
    - Mandatory: Yes (`*`)
    - Type: Email Input
    - Max. Length: 100
    - Default value: Empty
    - Validation: Must be a valid corporate email format (user@domain.com).
- **Department:**
    - Mandatory: No (Optional select)
    - Type: Dropdown (Single Select)
    - Default value: Business Analysis
    - Allowed Values: Business Analysis, Game, Collab, Web, Data analytics
- **Status:**
    - Mandatory: No (Optional select)
    - Type: Dropdown (Single Select)
    - Default value: Confirmed

    - Allowed Values: Confirmed, Intern, Probation, Notice Period
- **Designation:**
    - Mandatory: No
    - Type: Text Input
    - Default value: Empty
- **Reporting Manager:**
    - Mandatory: No
    - Type: Text Input
    - Default value: Empty
- **Auto-Generated Profile Defaults on Save:**
    - Company Mobile: +1 9005049999 (or +1 9887654321 via Auto-Fill)
    - Employee Code: Automatically generated
    - Joining Date: 04-Nov-2026
    - Account Status: Active
- **Form Actions:** Auto-Fill, Cancel, Save Employee.

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4. Global Training Configuration (Config > Training Categories)

General Settings Tab

- **Enable / Disable Training Module:**
    - Mandatory: Yes
    - Type: Toggle Switch (Active / Inactive)
    - Default value: Active
    - Description: Master switch to enable or disable the entire Training module access for employees across the portal.
- **Allow Training Creation:**
    - Mandatory: Yes
    - Type: Toggle Switch
    - Default value: Active
    - Description: Permit authorized roles to author and create new training programs.
- **Allow Training Editing:**
    - Mandatory: Yes
    - Type: Toggle Switch

- Default value: Active
- Description: Allow modification of existing published and draft training programs.

- **Allow Training Archiving:**
  - Mandatory: Yes
  - Type: Toggle Switch
  - Default value: Inactive
  - Description: Enable archiving of obsolete or retired training courses.
- **Enable Training Preview:**
  - Mandatory: Yes
  - Type: Toggle Switch
  - Default value: Active
  - Description: Allow creators and reviewers to preview interactive course modules.

Training Categories Management

- **Categories Table Columns:** Category Name, Description, Status, Created By & Date, Modified By, Actions (Edit icon + Active/Inactive toggle).
- **Pre-configured Categories (9 Default Entries):** Engineering & Architecture, Sales & Business Development, Recruitment & HR Operations, Security, Privacy & Compliance, Leadership & Management, Product Design & UX, Customer Success & Support, DevOps & Cloud Operations, Artificial Intelligence & ML.
- **Add Training Category Modal (+ Add Category):**
  - **Category Name:** Mandatory (`*`), Text Input (Alphanumeric). Validation: Case-insensitive unique check against all existing categories; duplicate category names are rejected with an alert.
  - **Description:** Optional Multi-line Textarea. Default value on save if blank: "New training category description."
  - **Status:** Mandatory, Toggle Switch, default Active.
  - **Quick Action:** Auto-Fill button populates sample category details.

Skills & Skill Levels Configuration (`Config > Skills`)

- **Configured Skill Levels (Ordered Progression Hierarchy):**
  - Stored centrally (`skillLevels` in `GET /api/config/skills`) and reused across Training Recognition and Employee Profiles without creating duplicate level masters:

    1. `Beginner` (Order: `1`)
    2. `Intermediate` (Order: `2`)
    3. `Advanced` (Order: `3`)

- **Skills Table Columns:** `Skill Name`, `Description`, `Status` (`Active` / `Inactive` toggle pill), `Created By`, `Actions` (`Edit`).
- **Add / Edit Skill Modal (`+ Add Skill`):**
  - **Skill Name:** Mandatory (`*`), Text Input. Validation: Case-insensitive uniqueness across master skills; duplicates rejected (`HTTP 400`).
  - **Description:** Optional Multi-line Textarea.
  - **Status:** Active / Inactive toggle (default Active). Only **Active** skills appear for selection in new Training configurations; deactivating a skill prevents future selection without affecting already-awarded Employee Profile skills.
- **Authorization:** Restricted to Admin, HR, and authorised Reporting Managers (`403 Forbidden` for Employee role).

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5. Role-Based Access Control (Config > Roles)

- **Roles Table Columns:** No., Role, Status, Employees, Remarks, Created By & Date, Modified By, Actions (View, Edit, Delete).
- **Standardized System Roles (11 Configured Roles):**
  1. Sr. Business Analyst (8 employees)
  2. Senior Recruiter (5 employees)
  3. CEO (3 employees)
  4. Project Manager (25 employees)
  5. BDE (1 employee)
  6. Admin/HR (50 employees)
  7. Employee (51 employees)
  8. Team leader (20 employees)
  9. Management (7 employees)
  10. Recruiter (1 employee)
  11. BDM (0 employees)
- **Add Role Slideover Modal (+ Add Role):**
  - **Title:** Mandatory (`*`), Text Input.
  - **Clone Role:** Optional Dropdown to copy permissions from an existing role.
  - **Remarks:** Optional Textarea, max length 255 characters.
  - **Status:** Toggle Switch, default Active.
- **Role Permission Matrix:**
  - Columns: Module, Status, View Default, Global, Add, Edit, Delete.

- Training Module permissions: Training Module (parent), Training Programs Library, Training Assignments, Employee Progress Tracking, Training Reviews & Evaluations, Certificate Generation & Issuance, Training Analytics & Reports.

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6. Certificate Verification & Credential Specifications

- **Certificate ID:** Structured format (`TRN-YYYY-Serial` , e.g. TRN-2026-0027, TRN-2026-0032).
- **Authorized Signatory:** Configured strictly as **Khushi Bhatt, Head of HR**.
- **Issued Date:** Day-Month-Year format (e.g. 27-Jan-2026).
- **Master-Detail Table Layout:**
  - Left columns: No. (1, 2, 3...), Employee Avatar & Name, Department / Role.
  - Middle column: Training Name (plain-text format, no dropdowns or passed badges).
  - Far-right column: Actions header and eye icon buttons aligned to the right directly beneath All Records.

SECTION 3: TRAINING LIFECYCLE & STATE TRANSITION FLOW

1. **Assigned**: Training is assigned to an Employee, Department, Designation, or Business Unit.
2. **In Progress**: Learner opens the assigned course and begins reviewing the learning resources.
3. **Completion Path (Assessment Disabled)**: Once 100% of mandatory learning modules (and any required document uploads) are completed, status transitions directly to **Completed** (or **Under Review** if `Require Reviewer / Manager Approval` is ON).
4. **Completion Path (Assessment Enabled)**: Learner submits the assessment (**Assessment Taken**).
   - **Pass**: When score meets/exceeds passing threshold and all other configured requirements are met, status transitions to **Completed** (or **Under Review** if reviewer approval is pending).
   - **Retake**: When score fails and attempts remain, status returns to **In Progress** for another attempt (no certificate or skill is awarded on fail).
5. **Reviewer / Manager Approval (`Under Review → Approved`)**: When `reviewerApprovalRequired` is ON (or `Get approval from the manager` is configured), an authorised Admin, HR, or Reporting Manager reviews and clicks **Approve Completion & Issue Recognition** (`POST /api/training/approve-completion`). Employees cannot approve their own completion (`HTTP 403`).

6. **Training Recognition Execution (Verified Completion Only)**: Once `verifyTrainingCompletionEligibility()` confirms all mandatory resources, assessment pass (if enabled), documents, and reviewer approval (if enabled) are complete:
   - If `Certificate Generation` is `ON`: generates authenticated certificate (`TRN-YYYY-Serial`) signed by **Khushi Bhatt, Head of HR**.
   - If `Add Skill to Employee Profile` is `ON`: assigns configured `selectedSkillIds` at configured `skillLevel` to the employee profile via duplicate-safe & level-upgrade logic and writes `skillAwardHistory` .
   - Both recognition mechanisms run completely independently (`Neither` , `Certificate Only` , `Skills Only` , or `Both`).
7. **Overdue**: If the target due date passes without completing modules or passing the assessment, status becomes **Overdue**.

SECTION 4: ASSESSMENT & PORTAL BEHAVIORAL SPECIFICATIONS

4.1 Assessment Toggle Behavior Matrix

System Component	Assessment Toggle: ON (Active)	Assessment Toggle: OFF (Inactive)
Tab 3 Header	3. Assessment & Questions	3. Assessment & Questions
Tab 3 Content	Renders full evaluation card, attempt counter, passing criteria, and all configured questions	Empty State Container: "No Assessment Configured" message with icon; no questions, no scoring inputs, no submit button
Left Sidebar Action Button	"Take Assessment →" (or "Download Certificate" when passed)	"Start Learning Modules →" (or "Programme Completed ✓" when all modules done)
Passing Score Display	80%	N/A (Assessment Disabled)
Tab 2 Navigation Button	"Proceed to Assessment →"	"All Modules Completed ✓"
Completion Criteria	All configured Completion Requirements satisfied (e.g. Videos + Passing Score on Assessment + Required Docs + Manager Approval)	All configured non-assessment Completion Requirements satisfied

4.2 Supported Question Formats

1. **Short Text Answer**: Single-line text input with case-insensitive accepted keyword evaluation.
2. **Long Text Answer**: Multi-line textarea for detailed subjective responses evaluated by reviewer.
3. **Single Selection (Radio)**: Radio group allowing one option selection.
4. **Multiple Selection (Checkbox)**: Checkbox group supporting multiple selections.
5. **Dropdown Selection**: Standard select menu with "-- Select Answer --" placeholder and option values.
6. **True / False**: Dual toggle buttons/radios for binary verification.

7. **Question Marks**: Every question displays its designated marks (the per-question Mandatory checkbox and tag have been removed from Create/Edit Programme, Programme Detail, and Employee Assessment).

4.3 Assignments Register Specifications

- **Scope Column Header**: **ASSIGNED SCOPE** (displays Employee, Department, or Designation assignment scope).
- **Total Record Count**: **41 Total** assignments.
- **Pagination**: Complete 5-page pagination (10 items per page with Previous/Next and page number controls).
- **Assignment Details View (`View Assignment`)**: Follows the two-column Employee Profile layout (Left learner sidebar with Status pill, Avatar, Name, Role, Department pill, Assigned Training, Assigned/Due dates; Right `General Info` tab with Top 3-box summary strip, `Assignment Information` grid, and `Assigned Learning Modules & Assessment` text rows).

4.4 Training Programmes Catalogue & Availability Status Architecture

- **Publication & Availability Status Rules (No `Archived` Status)**:
  - **Publication Status (`Status ↕`)**: Strictly **Published** or **Draft** (`Archived` status and archiving actions are removed).
  - **Published**: Emerald badge in the catalogue table. Visible on Management side; Active published programmes are assignable and usable on Employee side.
  - **Draft**: Amber badge. Management authoring only (Availability Status stays locked to Inactive until published and hidden from Employee side).
  - **Active / Inactive Only After Publish**: Once a programme is **Published**, only **Active** / **Inactive** transitions can be performed via the compact toggle switch placed at the top-right of the **Programme Detail** left sidebar card.
  - **Started Learner Lock (Deactivation Guard)**:
  - Once any employee has started working on a programme (In Progress, Under Review, Submitted for Review, Changes Requested, or Overdue with progress started), that programme cannot be made **Inactive**. Attempting deactivation is blocked with a validation error.
  - Once all learners have finished (Completed at 100%) or before any learner starts (Not Started at 0%), setting **Inactive** is permitted.
  - **Inactive Action Lock**:
  - **Management (Programme Detail)**: Displays an Inactive warning banner and disables `+ Assign Programme`, `Assign Programme`, `+ Add Resource`, and curriculum View action while Inactive. The sidebar availability toggle remains enabled for administration.

  - **Employee (My Training & Course Detail)**: Programmes completed first and then set to Inactive remain visible in My Training (with Completed + Inactive pills) in strict read-only mode; all interactive controls inside are disabled.
- **Catalogue Table Column Structure (7 Columns)**:
  1. **No.**
  2. **Training Name** ↕ (Programme title and version pill)
  3. **Level** ↕ (Beginner / Intermediate / Expert badge)
  4. **Duration** ↕ (Expected hours)
  5. **Enrolled** ↕ (Consistent enrolled learner count matching Programme Detail's Enrolled Employees list)
  6. **Status** ↕ (`Published` or `Draft` badge — no `Archived` or extra `Active` pill)
  7. **Actions** (Right-aligned View and Edit action buttons)
- **Programme Detail View (Management)**:
  - **Left Sidebar**: Displays publication badge (`Published` / `Draft`) on the top-left and the **Active / Inactive** toggle switch on the top-right, followed by Programme Title, Category, Difficulty Level, `Assign Programme` button, and core summary metrics (**Expected Duration**, **Passing Assessment Score** as clean `80%` , **Department**, **Enrolled Learners**).
  - **General Info Tab (Right Panel)**: Displays all programme details not shown on the left sidebar — **Learning Objectives**, **Completion Requirements & Assessment Criteria**, the **Training Recognition** summary card (Certificate Generation state, Add Skill to Employee Profile state, Reviewer Approval requirement, Awarded Skill Level, and configured Skills chips), and the **Assessment & Questions** section (Assessment Enabled/Disabled state, Assessment Title, Description, Total Marks, Max Attempts, and the full Questions List with question type, marks, and configured options/answers when present) cleanly without duplicating the sidebar summary details.
  - **Curriculum & Modules Tab**: Displays learning resources in a structured table (heading **Curriculum & Modules**, `+ Add Resource` button, and columns Module, Resource Title, Type, Resource URL, Action with eye View icon linking directly to the resource).
  - **Enrolled Employees & Status History Tabs**: Tracks enrolled learners (clean full-width learner search bar without extra `ENROLLED EMPLOYEES` count heading) with exact count consistency against the Training Management listing, plus audit history for `Active` / `Inactive` availability transitions (clean full-width status history search bar without redundant `Availability Status Audit History` title or `Programme ID` header).
  - **Uniform Employee Column Standard**: Across all employee-centric tables (Employees Directory, Assignments, Programme Detail > Enrolled Employees, Progress, and Certificates), the employee identity column uses the uniform **Employee Name** label (`Employee Name ↕` where sortable).

4.5 System Typography Standard

- **Uniform Sans-Serif Typography**: Every text element across the entire system (headings, tables, labels, duration metrics, version badges, IDs, percentages, URLs, modals, and certificates) consistently uses the exact same **Sans-Serif** font family (`--font-sans`), with no monospace or serif font overrides.

4.6 Training Recognition, Automatic Skill Assignment & Skill History Specifications

Recognition Combination Matrix

Certificate Generation	Add Skill to Employee Profile	Outcome on Verified Completion
OFF	**OFF**	Training marked Completed; **neither** certificate nor skills awarded
ON	**OFF**	Only **Certificate** issued; no skills added
OFF	**ON**	Only configured **Skills** added/upgraded on Employee Profile; no certificate issued
ON	**ON**	**Both** Certificate issuance and Skill assignment execute independently

Existing Skill Duplicate Handling & Level Progression Matrix

Before awarding any skill (`processTrainingRecognitionOnCompletion`), the backend checks the employee's canonical skill records (`canonicalEmployeeSkillKey`) and compares rank via `getSkillLevelRank` (`Beginner=1 < Intermediate=2 < Advanced=3`):

Existing Employee Level	Newly Awarded Level	Final Profile Level	Outcome & Record Behavior
(Not Present)	Any (e.g. `Beginner`)	`Beginner`	Creates single new skill record linked to source Training
`Beginner`	`Intermediate`	`Intermediate`	Updates existing record (`upgraded`); no duplicate created
`Intermediate`	`Advanced`	`Advanced`	Updates existing record (`upgraded`); no duplicate created
`Advanced`	`Beginner`	`Advanced`	Retains higher existing level (`retained`); **no downgrade**
`Intermediate`	`Intermediate`	`Intermediate`	Retains existing level (`retained`) and appends Training source

Skill Award History Audit Schema (`skillAwardHistory`)

Every skill award or level evaluation stores an immutable audit entry with:

1. `employeeId` (Canonical Employee Identifier, e.g. `CO-5039`)
2. `skillId` & `skillName`
3. `awardedSkillLevel`, `previousLevel`, `finalLevel`, `outcome` (`added` / `upgraded` / `retained`)
4. `sourceTrainingId` & `sourceTrainingName`
5. `awardedDate`, `createdDate`, `updatedDate`

Training Deactivation, Archive, Edit & Authorization Safeguards

- **Draft Progress Visibility**: Trainings with `Draft` publication status hide progress percentages on Progress/Training listings until published.
- **Preservation on Deactivation / Archival / Edit**: Deactivating, archiving, or editing a Training's recognition configuration after skills have been awarded **never** removes or downgrades previously awarded Employee Profile skills or `skillAwardHistory`.
- **Server-Side RBAC & Eligibility Enforcement**:
  - `POST /api/training/employee-course-progress`: Validates completion eligibility server-side (`verifyTrainingCompletionEligibility`) before issuing certificates or skills; rejects Employee self-approval (`managerApprovalStatus: Approved`) with `HTTP 403`.
  - `POST /api/training/approve-completion`: Authorised for Admin, HR, and Reporting Managers only (`HTTP 403` for Employees).

- `POST /api/config/skills` & `POST /api/employee/skills`: Authorised for Admin, HR, and Reporting Managers only (`HTTP 403` for Employees — employees cannot manually award or modify skills).